

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign Jun 2021 - present
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign Aug 2017 - May 2021

WORK EXPERIENCE

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483) Jun 2021 - present

- ❖ Taught lessons or held office hours for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing firmware, software, and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated weekly builds in TortoiseSVN using Python, reducing need for manual integration.

RESEARCH

Aerodynamics and Planetary Exploration Group (APEX) Advisor: Dr. Siegfried Eggli
Interstellar Navigation and Star Identification Sep 2024 - present

- ❖ Investigating feasibility of star identification (star-ID) at relativistic speeds and interstellar distances by combining traditional star-ID algorithms with clustering algorithms based on stellar classification.

Celestial Navigation to Supplement DSN Tracking of OSIRIS-APEX Jul 2024 - present

- ❖ Assessing feasibility of using celestial (asteroid) navigation to meet cruise phase uncertainty requirements.
- ❖ Designing sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- ❖ Implemented spherical harmonics for an open source orbit propagator to analyze cislunar missions.

Lost in Space and Time Navigation using Variable Stars Nov 2022 - present

- ❖ Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- ❖ Performed Monte Carlo analysis to conclude that reestablishing Earth contact via the proposed method using only commercial off-the-shelf (COTS) star trackers is possible.

Aerospace Mission Analysis Laboratory Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation Jun 2021 - present

- ❖ Developed an efficient algorithm that resolves spacecraft position and time from pulsar timing data.
- ❖ Created a photon time of arrival simulation tool in Python to synthesize the observed light curves of variable stars; validated results against the pulsar timing package [PINT](#) to within 50 μ s.
- ❖ Demonstrated the computational feasibility of X-ray pulsar navigation in a lost in space scenario.

Velocity-based Initial Orbit Determination

Aug 2019 - Apr 2024

- ❖ Developed a new initial orbit determination (IOD) technique using sequential range-rate measurements.
- ❖ Assessed the performance of various velocity-based IOD methods using Monte Carlo simulation.

Space Systems Optimization Laboratory

Advisor: Dr. Koki Ho

Architecture Trade Studies on IRSU Technologies for Human Space Exploration

Sep 2018 - Sep 2019

- ❖ Conducted literature review of in-situ resource utilization (ISRU) technologies to quantify input/output rates of technology options for an architecture optimization tool.

SKILLS

Programming Languages Python, C/C++, MATLAB

Software/Prepackaged Tools L^AT_EX, Git, GMAT

Operating Systems Linux, MacOS, Windows

Languages English, Mandarin Chinese

HONORS AND AWARDS

Yee Fellowship

2021, 2024

Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship

2024

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference

2024

American Astronautical Society (AAS) Rocky Mountain Section

David and Catherine Thompson Space Technology Scholarship

2020

American Institute of Aeronautics and Astronautics (AIAA)

H.S. Stillwell Problem Solving Award

2020

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Robert W. McCloy Memorial Award

2019

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

James Scholar

2018 - 2021

University of Illinois Urbana-Champaign

Dean's List

2017 - 2021

University of Illinois Urbana-Champaign

JOURNAL ARTICLES

⁵**L. Hou** et al., "Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars", *The Journal of the Astronautical Sciences*, in review (2024).

⁴**L. Hou** and Z. R. Putnam, "Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements", *The Journal of the Astronautical Sciences* **71**, 24 (2024).

- ³**L. Hou** and Z. R. Putnam, “A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV”, [The Journal of the Astronautical Sciences](#) **71**, 6 (2024).
- ²H. Chen et al., “Multifidelity Space Mission Planning and Infrastructure Design Framework for Space Resource Logistics”, [Journal of Spacecraft and Rockets](#) **58**, 538–551 (2021).
- ¹H. Chen et al., “Integrated In-Situ Resource Utilization System Design and Logistics for Mars Exploration”, [Acta Astronautica](#) **170**, 80–92 (2020).

CONFERENCE PROCEEDINGS

- ⁸**L. Hou** and S. Eggl, “Autonomous Optical Navigation using Astrometry and Photometry”, in AAS/AIAA Space Flight Mechanics Meeting, AAS 25-248 [accepted] (Jan. 2025).
- ⁷**L. Hou** et al., “Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars (AAS-24-012)”, in [46th Rocky Mountain AAS GN&C Conference](#) (Feb. 2024).
- ⁶A. Auster et al., “Mars Ice Thermal Harvesting Rig & ISRU Laboratory (MITHRIL)”, in [ASCEND 2022](#) (Aug. 2022), p. 4249.
- ⁵**L. Hou** and Z. R. Putnam, “A Norm-minimization Method for Solving the Cold-Start Problem with XNAV”, in AAS/AIAA Astrodynamics Specialist Conference, AAS 22-560 (Aug. 2022).
- ⁴**L. Hou** and Z. R. Putnam, “Initial Orbit Determination from Sequential Line-of-Sight Velocity Measurements”, in AAS/AIAA Astrodynamics Specialist Conference, AAS 22-561 (Aug. 2022).
- ³**L. Hou**, K. Lohan, and Z. R. Putnam, “Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms”, in AAS/AIAA Space Flight Mechanics Meeting, AAS 21-280 (Feb. 2021).
- ²H. Chen et al., “Multi-Fidelity Space Mission Planning and Space Infrastructure Design Framework for Space Resource Logistics”, in [AIAA Propulsion and Energy 2019 Forum](#) (2019), p. 4134.
- ¹H. Chen et al., “Integrated Analysis Framework for Space Propellant Logistics: Production, Storage, and Transportation”, [Lunar ISRU 2019-Developing a New Space Economy Through Lunar Resources and Their Utilization](#) **2152**, 5003 (2019).